

Who 'likes' alcohol? Young Australians' engagement with alcohol marketing via social media and related alcohol consumption patterns

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Alcohol consumption is an important public health issue in Australia; particularly high rates of consumption are observed across younger populations. Both risky episodic usage, and long-term frequent usage are associated with a plethora of harms, contributing significantly to morbidity, mortality and health care expenditure.¹ Alcohol has consistently been reported as one of the leading causes of disease and injury burden for young people. According to the 2013 National Drug Strategy Household Survey² one third of 18-24 year olds reported 'very high risk' alcohol consumption of 11 or more drinks in a single episode in the past year. Further, nearly 60% of young people report drinking with the aim of getting drunk in 2014.³ Previous research has established that heavy alcohol consumption is associated with a range of factors at the individual, cultural and socio-economic levels. Examples of consistent correlates include familial and peer influence, early initiation of drinking and living in a rural or remote area.⁴⁻⁶ Alcohol use is also considered to be strongly influenced by socio-cultural norms; these are shaped not only by direct social environments but also by media and popular culture. Of more recent concern is the role that social media can play in shaping norms, given its ubiquitous presence in young people's lives. Research by the Australian Communications and Media Authority⁷ found that 99% of 16-17 year olds had used a social networking site (SNS). The majority of SNS users are daily users, with the average young person aged 14-17 years spending about 2.5 hours per day on Facebook alone.⁸

Abstract

Objective: To describe patterns of 'liking' alcohol marketing social media pages, and determine related alcohol consumption patterns among young Australians.

Methods: Participants were 1,001 Australians aged 15-29 years who completed a cross-sectional online survey. Logistic regression and ordinal logistic regression were used.

Results: A quarter (249/1001, 24.9%) liked at least one of the alcohol marketing social media pages, most commonly brands of spirits, cider and alcohol retailers. Underage participants were as likely as older participants to report liking these pages. Alcohol marketing social media use was significantly and independently associated with male gender, living outside a major city, ever using illegal drugs and early age of first alcohol consumption (all $p < 0.05$). Alcohol marketing social media use (OR 2.1, 95% CI 1.5-2.8, $p = < 0.001$) was independently associated with higher categories on the AUDIT-C, indicating riskier alcohol consumption.

Conclusions: Liking or following alcohol marketing pages is common regardless of age, and associated with riskier alcohol consumption, among young Australians.

Implications: There is a need to develop strategies to reduce the exposure to, and potential impact of, alcohol marketing social media pages on young Australians, and ensure these pages are neither accessible to nor targeting underage social media users.

Key words: alcohol, young people, social media, marketing

Major alcohol companies are more frequently advertising via social media. From around 2010-2011, a surge in 'likes', 'shares' and posts to alcohol brands' Facebook pages was observed,⁹ and alcohol companies have been able to advertise on Twitter since 2011.¹⁰ Alcohol marketing on social media appears to be successful in terms of audience reach – according to a 2015 poll, 30% of young Australian adults have noticed alcohol advertising or promotions on social media.³ As of April 2016, the ten most popular alcohol brands worldwide have between 6.4 million and 20.7 million Facebook likes, 233,000-550,000 Twitter followers, and 11,000-190,000 YouTube subscribers.¹¹

Social media advertising allows alcohol companies to advertise in novel, interactive

ways. Alcohol companies may advertise to social media users via traditional-style banner advertisements which can be targeted to user demographics.¹² However, alcohol companies can also set up branded profiles or pages which allow the company to post text, image and video advertisements; these can be viewed by any internet user who 'clicks' on the brand's social media profile or who 'likes' or 'follows' the profile, automatically subscribing to updates, after which content from the company's profile will appear in the user's newsfeed. Users can simply view the content or they may choose to interact with the content via 'liking', 'commenting', 'sharing' and other forms of interaction, and this content may then appear in the newsfeeds of some of their friends in their social media

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Submitted: December 2015; Revision requested: April 2015; Accepted: May 2015

The authors have stated they have no conflict of interest.

networks. The amount of advertising material that appears in the user's newsfeed depends on their profile settings and the social media platform's algorithms. Alcohol companies typically use a number of advertising strategies to engage with their audience via social media, including promoting real-world tie-ins, interactive games and competitions, and time-specific suggestions to consume alcohol, and by facilitating brand-centric discussions with consumers.¹³ Nearly one third of young Australians who have seen alcohol marketing material on social media have reported interacting with an alcohol marketing page, such as by liking a page or 'checking in',³ which increases the page's reach to members of the user's social network. Alcohol brands on Facebook receive more engagement by post than any other industry.¹⁴

There is consistent evidence indicating that exposure to traditional alcohol advertising is associated with both earlier onset of alcohol consumption among non-drinkers and increased consumption among existing drinkers.¹⁵ Emerging research indicates that exposure to alcohol marketing via social media may also influence real life behaviours among young people. Jones et al.¹⁶ recently published cross-sectional research which found that, in a convenience sample of 283 Australian Facebook users aged 16-24 in 2013, 20.8% of respondents had visited an alcohol page on Facebook, and interaction with alcohol marketing was fairly common, with 87 participants (30.7% of the total sample) ever uploading or tagging a photo with an alcohol product, 17.7% liking something posted by an alcohol brand, and 13.4% viewing an event created or sponsored by an alcohol company. Their analysis found that respondents who interacted with alcohol marketing on Facebook were twice as likely to report more frequent alcohol consumption and 3.7 times more likely to report increased volume of alcohol consumption. Interacting with these pages – rather than exposure to more traditional alcohol advertisements on social media – was associated with hazardous alcohol consumption.¹⁶ In a US college sample, alcohol-related social media use has been associated with more frequent alcohol consumption in the last 30 days, higher alcohol consumption on a single occasion, and problems associated with alcohol consumption (e.g. social problems, injuries, financial problems).¹⁷ Experimental research has found that a greater interest in interacting with alcohol marketing on Facebook

is associated with greater intentions to consume alcohol among US college students, particularly when the message was paired with an already high rate of interaction from other Facebook users, regardless of safe consumption messages that may be present alongside advertisements.¹⁸

There is a need to further explore the relationship between alcohol marketing social media use and alcohol consumption in young people. Although Jones et al.¹⁶ studied a young Australian sample, they only studied Facebook-based alcohol marketing, their analysis was limited by a relatively small sample size, and their sample was derived from a market research panel which may have limited generalisability. Our study builds on this work by further exploring the relationship between alcohol marketing social media use and alcohol consumption among a large sample mainly comprised of social media users, with respect to types of alcohol marketing pages liked or followed, and inclusion of other risk behaviours (e.g. illegal drug use). Our study thereby aimed to describe patterns of liking and following different types of alcohol marketing social media pages, and determine whether liking or following these pages is associated with patterns of alcohol consumption, among a sample of adolescent and young adult Australians. It was hypothesised that liking or following alcohol marketing pages would be associated with higher rates of risky alcohol consumption.

Method

Data

Participants completed the 2015 *Sex, Drugs and Rock'n'Roll* study, an annual study involving a cross-sectional convenience sample of 15-29 year-old Victorians developed by the Burnet Institute.¹⁹ The study involved administration of a 112 question open online survey, which included questions on demographics, social media use, and alcohol use. Participants were recruited via social media, targeted Facebook advertisements and word of mouth. Informed consent was obtained from each participant via a plain language statement at the start of the online survey. The survey was open online for six weeks in February-March 2015. Approval for the *Sex, Drugs and Rock'n'Roll* study was granted by the Alfred Hospital Human Research Ethics Committee (326/08). No specific funding was received for this study.

Measures

Demographic variables

Demographic variables used included gender (male/female/transgender/other), age (15-17, 18-19, 20-29), country of birth (Australian-born, yes/no), highest level of completed education (currently completing or had completed any post-high school education, yes/no), sexual identity (heterosexual/gay, homosexual, lesbian, bisexual, queer, questioning or other; GLBQQ+), having \$120 or more for recreational spending on oneself per week (yes/no) and living in a major city (yes/no). Regionality (major city/other) was determined by participants' postal codes according to The Australian Standard Geographical Classification – Remoteness Area.²⁰

Alcohol marketing social media use

Participants were asked 'Do you like/follow any of the following types of pages on Facebook, Instagram or Twitter?' with the option to tick all that apply. The five options were as follows:

- Brand of cider (e.g. Rekorderlig, Kopparberg)
- Brand of beer (e.g. Carlton Dry, Corona)
- Brand of spirits (e.g. Smirnoff, Bundaberg)
- Brand of wine (e.g. Brown Brothers, Yalumba)
- Alcohol retailer (e.g. Dan Murphy's, Liquorland)

Binary variables were created to identify whether participants liked or followed each type of alcohol marketing page, whether participants liked or followed any of the alcohol brand pages (i.e. at least one of cider, beer, spirits or wine pages), and whether participants liked or followed any alcohol marketing pages in general (inclusive of alcohol retailers).

Alcohol consumption

Participants completed the AUDIT-C,²¹ a brief screening tool based on the Alcohol Use Disorders Identification Test (AUDIT).²² The three item AUDIT-C items asks the frequency of alcohol consumption, number of standard drinks consumed on a typical day of drinking, and risky single occasion drinking (defined in the AUDIT-C as consumption of six or more standard drinks on one occasion). The AUDIT-C is designed to identify heavy or hazardous alcohol consumption and/or potential alcohol use disorders, and we categorised scores to assess risk. Most

scoring systems for the AUDIT-C differentiate between women and men.²¹ However, as Australian alcohol consumption guidelines²³ no longer differentiate between male and female drinking patterns in determining risk, we used three categories of increasing risk for both men and women: 'lowest risk' (scores 0-3), 'medium risk' (scores 4-8) and 'highest risk' (scores 8-12).

Participants were also asked 'How old were you when you had your first full serve of alcohol? For example, a full serve refers to consuming a full drink on your own, as opposed to just having a taste of someone else's drink.' The question was accompanied by an image demonstrating 'standard drink' amounts for different alcohol types. Options for response included '<10' and each age between 10 and 29 years. This question was dichotomised around the median to inform 'early age of first alcohol consumption'.

Other variables

Participants were asked if they had experienced any mental health problems in the last six months (including any they hadn't spoken to a doctor about) and to self-rate their mental health on a 5 point scale from 'excellent' to 'poor'. Participants who answered either 'yes' to experiencing mental health problems, or self-rated their mental health as 'poor' or 'fair', were classified as having recent mental health problems (yes/no). Participants were also asked if they had ever used illegal drugs (yes/no).

Analysis

All statistical analyses were performed using Stata version 13. The first analysis involved cross-tabulations and univariable logistic regression to explore types of alcohol marketing social media pages liked or followed according to gender and age. Logistic regression was then used to determine variables associated with liking or following any alcohol marketing social media page without the influence of AUDIT-C scores. In the second analysis, ordinal logistic regression was used to determine variables associated with AUDIT-C scores. In both multivariable regression analyses, variables significant at $p < 0.05$ at the univariable level were included in multivariable models.

Results

The survey was completed by 1,001 participants; the median age was 21.1 years (IQR 17.6-24.6). Genders identified were

female (723; 72.2%), male (269; 26.9%), transgender (4; 0.4%), other (3; 0.3%) and two participants did not specify their gender. Most participants (898; 91.2%) reported ever consuming alcohol and the median age of first alcohol consumption was 15 years (IQR 14-17).

Analysis 1: Fans of alcohol marketing social media

A quarter of the sample (249/1001, 24.9%) liked or followed at least one of the alcohol marketing social media pages (Table 1). Of the brand types, the most frequently liked or followed on social media was spirits (14.1%) followed by cider (11.9%). In univariable logistic regression, male participants were significantly more likely than female participants to be fans of beer (OR 5.05, 95%CI 3.2-7.9), alcohol brands in general (OR 1.5, 95% CI 1.1-2.0), alcohol retailers (OR 1.6, 95% CI 1.1-2.3) and alcohol marketing pages in general (OR 1.5, 95% CI 1.1-2.1). Participants aged 15-17 years were significantly less likely than participants aged 20-29 years to be fans of brands of cider (OR 0.53, 95% CI 0.3-0.9). The vast majority of participants who liked or followed alcohol marketing pages had consumed alcohol (244; 99.6%).

In the multivariable model using logistic regression (pseudo $R^2 = 0.09$), liking or following any alcohol marketing page was significantly associated (at $p < 0.05$) with male gender, living outside of a major city, ever using illegal drugs and early age of first alcohol consumption (Table 2).

Analysis 2: Alcohol marketing social media use and alcohol consumption

Eight hundred and sixty-seven participants completed the AUDIT-C (which was only displayed to participants who had ever consumed alcohol, $n = 868$) and were included in analysis 2. The median score

on the AUDIT-C was 4 (IQR 3-6) and the mean was 4.6 ($SD = 2.5$). Using the three risk categories, 289 (33.3%) scored 0-3, indicating lowest risk drinking; 458 (52.8%) scored 4-7, indicating medium risk drinking; 120 (13.8%) scored 8-12, indicating highest risk drinking.

Table 3 shows that in the multivariable model (pseudo $R^2 = 0.15$), higher categories on the AUDIT-C (indicating more risky alcohol consumption) were associated with liking or following alcohol marketing pages (OR 2.06, 95% CI 1.51-2.82, $p = < 0.001$). Higher categories on the AUDIT-C were also associated with older age, being born in Australia, more money for recreational spending per week, ever using illegal drugs, and younger age at first alcohol consumption.

Discussion

This cross-sectional study aimed to describe patterns of liking and following alcohol marketing social media pages, and determine whether liking or following these pages was associated with risky alcohol consumption, among a sample of young Australians. Around one quarter of the sample liked or followed at least one of the alcohol marketing pages. Liking or following alcohol marketing pages was independently associated with male gender, residing outside of a major city, and younger age of first alcohol consumption. Participants who liked or followed alcohol marketing pages were twice as likely as those who did not like the pages to report risky alcohol consumption (i.e. higher than 'lowest risk' AUDIT-C scores) which was independently associated with male gender, older age, being born in Australia, higher recreational spending money, ever using illegal drugs, and younger age of first alcohol consumption. The relationship between liking or following alcohol marketing social media pages and risky alcohol consumption

Table 1: Alcohol marketing social media fans by type of page, gender and age.

Type of page	Gender		Age			Total fans N(%)
	Male n(%)	Female n(%)	15-17 n(%)	18-19 n(%)	20-29 N(%)	
Any alcohol brand	73 (27.1) ^c	146 (20.2)	55 (19.7)	27 (21.1)	140 (23.6)	222 (22.2)
Brand of cider	31 (11.5)	86 (11.9)	22 (7.9) ^a	15 (11.7)	82 (13.8)	119 (11.9)
Brand of beer	55 (20.5) ^c	35 (4.8)	21 (7.5)	9 (7.0)	61 (10.3)	91 (9.1)
Brand of spirits	43 (16.0)	97 (13.4)	47 (16.9)	22 (17.2)	72 (12.1)	141 (14.1)
Brand of wine	11 (4.1)	41 (5.7)	10 (3.6)	5 (3.9)	38 (3.9)	53 (5.3)
Alcohol retailer	47 (17.5) ^c	85 (11.8)	39 (14.0)	14 (10.9)	81 (13.6)	134 (13.4)
Any alcohol marketing page	83 (30.9) ^b	163 (22.5)	63 (22.6)	30 (23.4)	156 (26.3)	249 (24.9)

a: $p < 0.05$, b: $p < 0.01$, c: $p < 0.001$

NB. Participants who did not identify as 'male' or 'female' were not included in the gender analysis.

supported our hypothesis and is consistent with emerging research.¹⁶⁻¹⁸

Liking or following alcohol marketing pages appeared common in our sample, at a much higher rate than that observed by in a recent Australian poll (6.1% of Australian adults)³ but only slightly higher than that of Jones et al. (20.8% of young Australians)¹⁶ who used a sample of Australians of similar age to ours. Our finding that over a quarter of young Australians in our sample are engaged with alcohol marketing highlights the degree to which young Australians are willingly exposing themselves to alcohol marketing content. Several factors may underpin this relationship. Firstly, young people may simply like and follow alcohol marketing pages as a signifier of brand interest or loyalty, establishing their favourite alcohol brands

and retailers.^{13,24} Secondly, it is possible that some young people are simply drawn to the alcohol marketing page for other reasons such as interest in a promotion, event or competition run by the brand or company. Young people may then continue liking or following the alcohol marketing page if the content is nonintrusive or appealing. Third, alcohol use is largely treated as a socially desirable activity for young Australians, and social media offers a mode of social approval, connection to others, and influence on identity.^{25,26} Therefore, publically interacting with alcohol marketing social media pages may be socially rewarded in this manner. A social networks analysis would be useful to study relationships between participants and their friends in terms of alcohol marketing social media use and alcohol consumption.²⁷

Gender and age differences were observed in the data, which were not observed in Jones et al.'s Australian study which utilised a similar, albeit smaller, sample.¹⁶ In our study, male participants were more likely than female participants to like or follow alcohol marketing pages (including beer brands, alcohol brands in general, and alcohol retailers), and more likely to report risky alcohol consumption on the AUDIT-C. This relationship is likely influenced by the normalisation of alcohol consumption for young men, which has been observed in relation to construction of masculine identity,²⁸ and generally higher rates of alcohol consumption and alcohol-related harms observed among young Australian men compared to young women.^{29,30} Further, the finding that living outside of a major city was associated with liking or following alcohol marketing pages is of interest. Rurality is consistently a risk factor for hazardous alcohol consumption,⁴ it is likely that this association reflects this trend.

Participants aged 15-17 were equally as likely as older participants to like or follow alcohol marketing pages, despite social media restrictions designed to prevent underage participants from accessing alcohol marketing on social media. These restrictions vary between social media sites; on Facebook, for example, official alcohol marketing pages and advertisements are only allowed to be visible to users over the legal alcohol consumption age, varying by country,^{31,32} while on Twitter users are required to enter their date of birth in order to follow an alcohol brand.³³ Therefore, in theory, official alcohol marketing pages should only be visible to social media users aged 18 years or older in Australia. However, our findings that underage users access alcohol marketing content on social media are consistent with previous studies of Facebook, Twitter and YouTube^{9,34,35} indicating that these age verification measures are not preventing underage users from accessing alcohol marketing pages. It is possible that some of our underage participants were referring to unofficial (i.e. user-led) alcohol marketing social media pages which are common¹² and are not subject to the same restrictions, falsifying their birthdates when setting up social media profiles, or not responding to the question accurately.

Alcohol marketing on social media provides significant challenges for regulatory bodies. Social media alcohol advertisements are subject to similar rules as alcohol

Table 2: Associations between liking or following any alcohol marketing social media page and other variables using logistic regression. Variables significant at $p < 0.05$ in univariable analyses were included in the multivariable model.

Variable	Total N(%)	Fan of alcohol social media page		Univariable			Multivariable		
		Yes n(%)	No n(%)	OR	95% CI	p	aOR	95% CI	p
Gender									
Female	723 (72.2)	163 (22.5)	560 (77.5)	1.0			1.0		
Male	269 (26.9)	83 (30.9)	186 (69.1)	1.5	1.1-2.1	0.007	1.6	1.1-2.2	0.007
Age									
15-17	279 (27.9)	63 (22.6)	216 (77.4)	0.9					
18-19	128 (12.8)	30 (23.4)	98 (76.6)	0.9	0.6-1.1	0.242			
20-29	594 (59.3)	156 (26.3)	438 (73.7)	1.0	0.5-1.3	0.508			
Education									
High school or lower	336 (22.6)	78 (23.1)	258 (76.8)	1.0					
Post high-school	664 (66.3)	171 (25.8)	493 (74.3)	1.4	0.8-1.6	0.381			
Location									
Major city	846 (84.5)	201 (23.8)	646 (76.2)	1.0			1.0		
Other area	137 (13.7)	44 (32.1)	93 (67.9)	1.5	1.0-2.5	0.037	1.6	1.0-2.4	0.033
Sexual identity									
Heterosexual	768 (76.7)	195 (25.4)	573 (74.6)	1.0					
GLBQQ+ ^a	229 (22.9)	54 (23.6)	175 (76.4)	0.9	0.6-1.3	0.579			
County of birth									
Outside Australia	97 (9.7)	22 (22.7)	75 (77.3)	1.0					
Australia	893 (89.2)	224 (25.1)	669 (74.9)	1.1	0.7-1.9	0.603			
Recreational spending per week									
<\$120 AUD	770 (76.9)	184 (23.9)	586 (76.1)	1.0					
\$120+ AUD	226 (22.6)	64 (28.3)	162 (71.7)	1.3	0.9-1.8	0.177			
Recent mental health problems									
Yes	566 (56.5)	141 (24.9)	425 (75.1)	1.0	0.7-1.3	0.992			
No	425 (42.5)	106 (24.9)	319 (75.1)	1.0					
Ever used illegal drugs									
Yes	545 (54.5)	173 (31.7)	372 (68.3)	2.4	1.7-3.2	<0.001	1.4	1.0-2.0	0.033
No	446 (44.6)	73 (16.4)	373 (83.6)	1.0			1.0		
Age at first alcohol consumption^b									
10-14	340 (34.5)	130 (38.2)	210 (61.8)	2.4	1.8-3.3	<0.001	2.2	1.6-3.0	<0.001
15+	558 (56.7)	114 (20.4)	444 (79.6)	1.0			1.0		

a: Gay, lesbian, bisexual, queer, questioning or other

b: Based on median age of first alcohol consumption in this sample (15 years).

N.B. Participants who did not identify as 'male' or 'female' were not included in the gender analysis

advertisements in other areas,³⁶ and social networking sites have their own advertising regulations. In Australia, standards for alcohol advertising are set by the Alcoholic Beverages Advertising Code (ABAC) and are designed to ensure alcohol is advertised responsibly.³⁷ Signatories to the Code must ensure that marketing communications do not show or encourage excessive alcohol consumption, strongly or evidently appeal to, or depict, minors, or suggest that the consumption of an alcoholic beverage creates or contributes to a change in mood or environment, relaxation, success or achievement. Our results suggest that alcohol marketing pages are appealing to minors. Further, a recent analysis indicates that, despite these

regulations, young people perceive that posts by alcohol companies on Facebook link alcohol to mood enhancement, and social and/or sexual success;³⁸ these factors may contribute towards their appeal. If regulatory bodies wish to update advertising standards in response to the results of our study and other emerging research, they must do so with consideration of the above points and the challenges of social media marketing and regulation.

Limitations

Our sample was an online convenience sample, generally reported high levels of risky alcohol consumption and was mostly recruited via social media; thereby results

may not be generalisable to other young Australians. Our sample included a high proportion of participants reporting GLBQQ+ identities, mental health problems and illegal drug use; although we controlled for these in multivariable analyses, it is possible these variables reflected other risk factors for alcohol consumption in our sample. The study was cross-sectional, thus we are unable to infer causality. This study only looked at liking or following alcohol marketing pages, a fairly simple measure of exposure to alcohol marketing via social media, and did not explore other interactions with these pages, which have been shown to be associated with consumption patterns¹⁶ and intention to consume alcohol in young adults.¹⁸ Data were

Table 3: Associations between AUDIT-C scores and other variables using ordered logistic regression. Variables significant at $p < 0.05$ in univariable analyses were included in the multivariable model.

Variable	Total N(%)	AUDIT-C scores			Univariable			Multivariable		
		0-3 (lowest risk) n(%)	4-7 (medium risk) n(%)	8-12 (highest risk) n(%)	OR	95% CI	<i>p</i>	aOR	95% CI	<i>p</i>
Gender										
Female	625 (72.1)	226 (36.2)	340 (54.4)	59 (9.4)	1.0			1.0		
Male	236 (27.2)	62 (26.3)	113 (47.9)	61 (25.9)	2.1	1.6-2.9	<0.001	1.8	1.3-2.5	<0.001
Age										
15-17	203 (23.4)	103 (50.7)	84 (41.4)	16 (7.9)	1.0			1.0		
18-19	116 (13.4)	38 (32.8)	59 (50.9)	19 (16.4)	2.2	1.4-3.5	<0.001	2.3	1.6-5.2	0.001
20-29	548 (63.2)	148 (27.0)	315 (57.5)	85 (15.5)	2.6	1.9-3.6	<0.001	2.6	1.5-4.8	0.002
Education										
High school or lower	253 (29.2)	112 (44.3)	112 (44.3)	29 (11.5)	1.0			1.0		
Post high-school	613 (70.7)	117 (28.9)	345 (56.3)	91 (14.9)	1.8	1.4-2.4	<0.001	0.9	0.6-1.6	0.812
Location										
Major city	741 (85.5)	247 (33.3)	390 (52.6)	104 (14.0)	1.0					
Other area	114 (13.2)	39 (34.2)	59 (51.8)	59 (16 (14.0)	1.0	0.7-1.4	0.885			
Sexual identity										
Heterosexual	664 (76.6)	223 (33.6)	347 (52.3)	94 (14.2)	1.0					
GLBQQ+ ^a	201 (23.2)	65 (32.3)	111 (55.2)	25 (12.4)	1.0	0.7-1.4	0.986			
County of birth										
Outside Australia	86 (9.9)	41 (47.7)	35 (40.7)	10 (11.6)	1.0			1.0		
Australia	773 (89.2)	244 (31.6)	419 (54.2)	110 (14.2)	1.8	1.2-2.8	0.007	2.5	1.5-4.0	<0.001
Recreational spending per week										
<\$120 AUD	650 (75.0)	245 (37.7)	338 (52.0)	67 (10.3)	1.0			1.0		
\$120+ AUD	214 (24.7)	44 (20.6)	118 (55.1)	52 (24.3)	2.5	1.9-3.5	<0.001	1.8	1.3-2.5	0.001
Recent mental health problems										
Yes	497 (57.3)	173 (34.8)	259 (52.1)	65 (13.1)	0.9	0.7-1.1	0.224			
No	363 (41.9)	113 (31.1)	196 (54.0)	54 (14.9)	1.0					
Ever used illegal drugs										
Yes	526 (60.7)	98 (18.6)	325 (61.8)	103 (18.6)	5.5	4.1-7.3	<0.001	4.4	3.2-6.0	<0.001
No	334 (38.5)	188 (56.3)	130 (38.9)	16 (4.8)	1.0			1.0		
Age at first alcohol consumption^b										
10-14	332 (38.3)	72 (21.7)	199 (59.9)	61 (18.4)	2.2	1.7-2.9	<0.001	1.6	1.2-2.2	0.002
15+	535 (61.7)	217 (40.6)	259 (48.4)	59 (11.0)	1.0			1.0		
Likes alcohol related social media										
Yes	242 (27.9)	51 (21.1)	134 (55.4)	47 (23.6)	2.6	1.8-3.3	<0.001	2.1	1.5-2.8	<0.001
No	625 (72.1)	238 (38.1)	324 (51.8)	63 (10.1)	1.0			1.0		

a: Gay, lesbian, bisexual, queer, questioning or other

b: Based on median age of first alcohol consumption in this sample (15 years)

N.B. The test of the proportional odds assumption using a likelihood ratio test was non-significant, as was the Brant Test of parallel regression. Participants who did not identify as 'male' or 'female' were not included in the gender analysis

self-reported and vulnerable to recall bias as social media users often like or follow a large number of pages and are unlikely to recall all pages. We only asked participants if they liked or followed alcohol marketing across Facebook, Instagram or Twitter, potentially excluding participants who liked these pages across other social media platforms such as YouTube. The AUDIT-C is a self-report screening tool (also subject to recall bias) and does not provide as rich an indication of alcohol related harms as the full AUDIT.²² Further, our definition of risky single-occasion drinking (6+ standard drinks) was drawn from the AUDIT-C and aligns with old Australian criteria for this drinking pattern rather than current criteria (4+ standard drinks).²³ Finally, our mental health measure was only a brief, single-item measure developed by the researchers, and has not been formally validated.

Conclusions and implications

Our findings indicate that liking or following alcohol marketing pages is common and associated with riskier alcohol consumption in our sample of young Australians. Further, underage participants report similar rates of liking or following alcohol marketing pages as older participants, and male participants reported both higher rates of liking or following the pages and engaging in more risky alcohol consumption. These results indicate that alcohol marketing on social media is reaching large numbers of young Australians despite current restrictions and regulations in place by advertising boards and social media platforms. The results of our study emphasise the need to further research strategies by social media moderators and alcohol regulation bodies to reduce the exposure to and potential impact of alcohol marketing pages on adolescents and young adults, and ensure these pages are neither accessible to nor targeting under-age social media users. Future research endeavours should consider social network analyses to explore alcohol marketing engagement among online friendship groups, and consider longitudinal study designs to provide more rigorous evidence.

References

- Chikritzts T, Pascal R. *Trends in Youth Alcohol Consumption and Related Harms in Australian Jurisdictions, 1990–2002*. Bulletin 8. Perth (AUST): Curtin University National Drug Research Institute; 2004.
- Australian Institute of Health and Welfare. *National Drug Strategy Household Survey Detailed Report 2013*. Canberra (AUST): AIHW; 2014.
- Foundation for Alcohol Research and Education. *Annual Alcohol Poll: Attitudes and Behaviours* [Internet]. Melbourne (AUST): Deakin University; 2015 [cited 2016 Aug 16]. Available from: <http://www.fare.org.au/alcpoll2015/>
- Miller PG, Coomber K, Staiger P, Zinkiewicz L, Toubmourou JW. Review of rural and regional alcohol research in Australia. *Aust J Rural Health*. 2010;18(3):110–17.
- Grant BF, Dawson DA. Age at onset of alcohol use and its association with DSM-IV alcohol abuse and dependence: Results from the national longitudinal alcohol epidemiologic survey. *J Subst Use*. 1997;9:103–10.
- Ary DV, Tildesley E, Hops H, Andrews J. The influence of parent, sibling, and peer modeling and attitudes on adolescent use of alcohol. *Int J Addict*. 1993;28(9):853–80.
- Australian Communications and Media Authority. *Like, Post, Share: Young Australians' Experience of Social Media*. Sydney (AUST): ACMA; 2013.
- Raco E. *Aussie Teens Online*. Sydney (AUST): Australian Communications and Media Authority; 2014.
- Jernigan DH, Rushman AE. Measuring youth exposure to alcohol marketing on social networking sites: Challenges and prospects. *J Public Health Policy*. 2014;35(1):91–104.
- Dugan L. *Once Banned, Alcohol Brands are Now Buying Ads* [Internet]. Twitter SocialTimes. 2011 [cited 2015 Oct 20]. Available from: <http://www.adweek.com/socialtimes/once-banned-alcohol-brands-are-now-buying-ads-on-twitter/449530>
- Socialbakers. *Facebook Brands Stats – Alcohol* [Internet]. New York (NY): Socialbakers; 2015 [cited 2016 Apr 18]. Available from: <http://www.socialbakers.com/statistics/facebook/pages/total/brands/alcohol/>
- Mart SM, Mergendoller J, Simon M. Alcohol promotion on Facebook. *J Glob Drug Policy Pract*. 2009;3:1–8.
- Nicholls J. Everyday, everywhere: alcohol marketing and social media - current trends. *Alcohol Alcohol*. 2012;47(4):486–93.
- Socialbakers. *Alcohol Brands Shake up Engagement* [Internet]. New York (NY): Socialbakers; 2012 [cited 2015 Nov 9]. Available from: <http://www.socialbakers.com/blog/1073-alcohol-brands-shake-up-engagement>
- Anderson P, de Bruijn A, Angus K, Gordon R, Hastings G. Impact of alcohol advertising and media exposure on adolescent alcohol use: A systematic review of longitudinal studies. *Alcohol Alcohol*. 2009;44(3):229–43.
- Jones SC, Robinson L, Barrie L, Francis K, Lee JK. Association between young Australian's drinking behaviours and their interactions with alcohol brands on Facebook: Results of an online survey. *Alcohol Alcohol*. 2016;51(4):474–80.
- Hoffman EW, Pinkleton BE, Weintraub Austin E, Reyes-Velazquez W. Exploring college students' use of general and alcohol-related social media and their associations with alcohol-related behaviors. *J Am Coll Health*. 2014;62(5):328–35.
- Alhabash S, McAlister AR, Quilliam ET, Richards JI, Lou C. Alcohol's getting a bit more social: When alcohol marketing messages on Facebook increase young adults' intentions to imbibe. *Mass Commun Soc*. 2015;18(3):350–75.
- Lim MS, Hellard ME, Aitken CK, Hocking JS. Sexual-risk behaviour, self-perceived risk and knowledge of sexually transmissible infections among young Australians attending a music festival. *Sex Health*. 2007;4(1):51–6.
- Department of Health. *Australian Standard Geographical Classification - Remoteness Area (ASGC-RA)* [Internet]. Canberra (AUST): Government of Australia; 2006 [cited 2016 Apr 26]. Available from: <http://www.doctorconnect.gov.au/internet/otd/publishing.nsf/content/ra-intro>
- Bush K, Kivlahan DR, McDonell MB, Fihn SD, Bradley KA. The AUDIT alcohol consumption questions (AUDIT-C): An effective brief screening test for problem drinking. Ambulatory Care Quality Improvement Project (ACQUIP). Alcohol Use Disorders Identification Test. *Arch Intern Med*. 1998;158(16):1789–95.
- Babor TF, Higgins-Biddle JC, Saunders JB, Monteiro MG. *AUDIT: The Alcohol Use Disorders Identification Test. Guidelines for Use in Primary Care*. Geneva (CHE): World Health Organization; 2001 [cited 2015 Nov 9]. Available from: http://www.talkingalcohol.com.au/files/pdfs/WHO_audit.pdf
- National Health and Medical Research Council. *Australian Guidelines to Reduce Health Risks from Drinking Alcohol*. Canberra (AUST): NHMRC; 2009.
- Nielson Group. *Tried and True or Adventurous: Do Consumers Stick to their Favorite Alcohol Brands?* [Internet]. New York (NY): Nielson; 2015 [cited 2015 Nov 9]. Available from: <http://www.nielson.com/us/en/insights/news/2015/tried-and-true-or-adventurous-most-consumers-stick-to-their-favorite-alcohol-brands.html>
- Ridout B, Campbell A, Ellis L. 'Off your Face(book)': Alcohol in online social identity construction and its relation to problem drinking in university students. *Drug Alcohol Rev*. 2012;31(1):20–6.
- Moreno MA, Kota R, Schoofs S, Whitehill JM. The Facebook influence model: A concept mapping approach. *Cyberpsychol Behav Soc Netw*. 2013;16(7):504–11.
- Otte E, Rousseau R. Social network analysis: A powerful strategy, also for the information sciences. *J Inform Sci*. 2002;28(6):441–53.
- de Visser RO, Smith JA. Alcohol consumption and masculine identity among young men. *Psychol Health*. 2007;22(5):595–614.
- Australian Bureau of Statistics. *4102.0 - Australian Social Trends, 2008* [Internet]. Canberra (AUST): ABS; 2008 [cited 2015 Nov 9]. Available from: <http://www.abs.gov.au/AUSSTATS/abs@.nsf/Lookup/4102.0Chapter5002008>
- Livingston M, Room R. Variations by age and sex in alcohol-related problematic behaviour per drinking volume and heavier drinking occasion. *Drug Alcohol Depend*. 2009;101(3):169–75.
- Facebook. *Advertising Policies - Restricted Content* [Internet]. Menlo Park (CA): Facebook; 2015 [cited 2015 Oct 22]. Available from: www.facebook.com/policies/ads/#restricted_content
- Facebook. *Facebook Pages Terms* [Internet]. Menlo Park (CA): Facebook; 2015 [cited 2015 Oct 22]. Available from: www.facebook.com/page_guidelines.php
- Jain T. *Age-screening Improvement for Alcohol Brands* [Internet]. San Francisco (CA): Twitter; 2013 [cited 2016 Aug 16]. Available from: <https://blog.twitter.com/2013/age-screening-improvement-for-alcohol-brands>
- Barry AE, Johnson E, Rabre A, Darville G, Donovan KM, Efunbunmi O. Underage access to online alcohol marketing content: A YouTube case study. *Alcohol Alcohol*. 2015;50(1):89–94.
- Winpenny EM, Marteau TM, Nolte E. Exposure of children and adolescents to alcohol marketing on social media websites. *Alcohol Alcohol*. 2014;49(2):154–9.
- Advertising Standards Bureau. *Social Media Advertising* [Internet]. Canberra (AUST): The Board; 2016 [cited 2016 Apr 26]. Available from: <https://adstandards.com.au/products-issues/social-media-advertising>
- Alcoholic Beverages Advertising Code. *ABAC Responsible Alcohol Marketing Code* [Internet]. Stirling (AUST): ABAC; 2014 [cited 2015 Nov 9]. Available from: <http://www.abac.org.au/wp-content/uploads/2014/06/ABAC-Responsible-Alcohol-Marketing-Code-30-4-14.pdf>
- Weaver E, Wright C, Dietze P, Lim MSC. "A drink that makes you feel happier, relaxed and loving": young people's perceptions of alcohol advertising on Facebook. *Alcohol Alcohol*. 2016;51(4):481–6.